

Meeting Summaries of October 30

Date: October 30

Time: 8:00–9:00 PM(PDT)

Attendees: Dr. Jan and Feng

Topic: Feng's work on the weight-driven PCA method for pollution assessment

Summary of Discussion

- Feng gave a brief summary of the work done last week, including:
 - A refactored codebase has been built, a package called 'zci' contains five major modules to implement the research objectives. The reported work is code released by the module - "sediment_pollution_assessment".
[Click here to access the GitHub repository](#), a detailed README file is provided there.
 - Feng talked about the views and reasons to inspire this idea, making a comparison with the benchmark PCA results and a weight-driven PCA results to show the differences in assessing the observed objects.

An analogy (Figure 2) of "standing at different views to observe objects to maximizingly distinguish them" is used to illustrate the idea of weight-driven PCA.

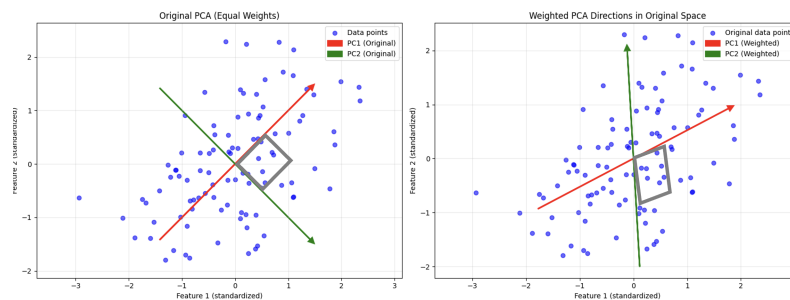


Figure 1: In the original data space, the selected objects changed when PCs are shifted by weights on features. Some were removed, some new ones were added and some remained.

- Prof. Jan pointed out that distinguishing the sites in pollution term maybe not the top priority. Recalling the main objective - 'zoobenthic community indicator of sediment pollution', how the sediment creatures respond to the contaminations should be the focus and identifying the factor(s) driving the community changes is more important for later work.

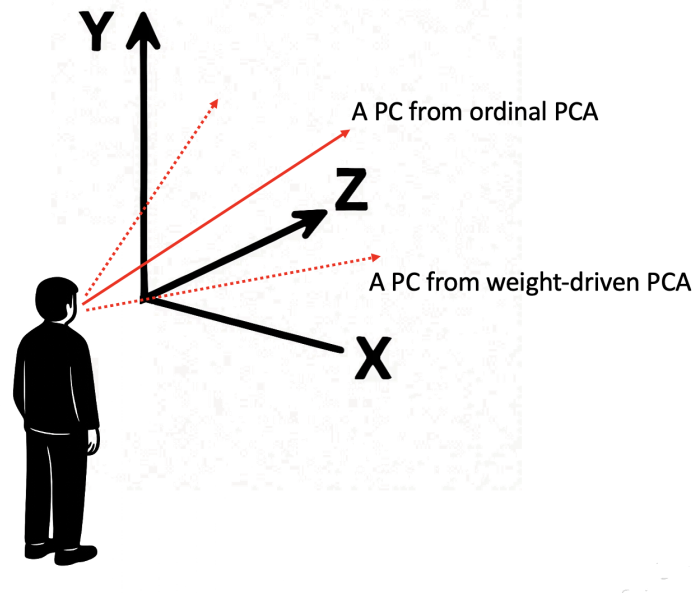


Figure 2: An analogy of "standing at different views to observe objects to maximizingly distinguish them".

My thinkings about it: This is a very critical and insightful point. I should not only focus on how to find the most polluted sites, but also how to identify the main stressor(s) or factor(s) driving the community changes under a certain pollution level (or pollution gradient).

Some factors/stressors mainly contribute to a site being highly polluted from a human perspective, but the benthic creatures might not respond to it noticeably. On the contrary, some stressors/factors contribute less to the assessed pollution level, but the benthic creatures are more sensitive to them and respond significantly.

It is hard to guarantee that the most pollution causing factors from human perspective are also the most influential factors to the benthic community changes. But in order to 'use' the benthic community as an indicator of sediment pollution for human purposes, we need to stand at the **benthic creatures' perspective** and balance the two kinds of views(human's and benthic creatures') in defining "what is an influential pollution".

Therefore, in the future work, I should consider how to connect the pollution assessment with the benthic community changes. It is not only building a regression model between the "pollution level" and the "benthic community metric(s)", but we can **let the "creatures" come into the pollution assessment stage** and help us to find an equilibrium point that measures the "pollution level" and "community changes" together.

- Feng talked about the metrics to measure what is a good weight to produce qualified PCs for pollution assessment. These good weight-driven PCs are like the views (as in Figure 2) that can both distinguish the sites by considering all features and shift the focus to the important features (high weight ones).

The two metrics are currently named as "Completeness Metric" and "Discrimination Metric".

- Feng introduced the 3 algorithms to achieve the above tasks.
 - Algorithm 1: It computes the shifted PCs and makes a selection criteria to choose the pollution-driven PC among them. It also computes the composite pollution scores by assigning new weights to the produced PCs ¹.

¹Here the weights of PCs are not set as input, but computed by the loading results of the selected pollution-driven PC.

- Algorithm 2: It evaluates the completeness of the selected weight-driven PCs. The algorithm receives the results from Algorithm 1 and fits a baseline PCA results without weights on stressors. Then qualified baseline PCs will be selected with the same steps in Algorithm 1 (it can be found in appendix). With the two groups of PCs, it computes the similarity between weight-driven PCs and the baseline PCs by pearson correlation coefficients. Additionally, multivariate linear regression models - X PCs are built with them and their R^2 is compared to evaluate how much information is preserved in the weight-driven PCs.
- Algorithm 3: It evaluates the discrimination of the selected weight-driven PCs. It receives the results from Algorithm 1 and produces the same baseline PCA results as in Algorithm 2, and it picks up the reference sites from the two groups of PCs. Using the two groups of reference sites, it makes hypothesis tests on the high-weight features and normal(low)-weight features:
 - * To the high-weight features:
Make hypothesis tests to check whether the weight-driven reference sites have lower means in these features than the baseline reference sites.
We want to see these feature means are significantly lower in the weight-driven reference sites.
 - * To the normal(low)-weight features: Make hypothesis tests to check whether there is no significant difference between the weight-driven feature means and the baseline feature means.
We want to see these weight-driven feature means are not significantly different from the baseline feature means.

Decisions Made

- Algorithms are not bad and can be good ideas but do not forget the main objective - 'zoobenthic community indicator of sediment pollution'.
- Even though the current work is inspiring and interesting, but it is still a part of the whole project. Therefore, it might be the time that the project work should move forward to the next stage (cluster analysis on the reference sites) and to see what is ahead.

Action Items

- Feng will complete and polish these ideas clearer and express them in a more acceptable and attractive way. Besides, the algorithm results should be tested under various scenarios (different weight settings, different input data, etc.) and the results should be presented in a more readable way (plots, tables, etc.).

Follow-Up

Temporary meeting time: November 6th, 16:00–16:45 PM(PDT)

Supportive materials

- A meeting recording with subtitles has been restored on Google Drive.

1 Appendix

1.1 Algorithms discussed in the meeting

1. **Weighted object:** directly multiplied by the weights, its inner value structure is reshaped by the weights.
2. **Weight-driven object:** indirectly influenced by the weights, its inner value structure is driven by the weights but not directly multiplied by the weights.

Algorithm 1: PCA-based Contamination Assessment with weights on stressor and qualified PCs selection

Data: Standardized Chemical data matrix $X \in \mathbb{R}^{n \times p}$, weights $\tilde{w} \in \mathbb{R}^p$, high-weighting threshold w^* ,
Result: Pollution scores $S_{\tilde{w}} \in \mathbb{R}^n$, loadings of selected PCs $Q_{sub|\tilde{w}}$

- 1 **Step 1:** Keep the high-weighted ($\tilde{w}_i > w^*$) stressors in the set I_{hw}
- 2 **for** stressor feature $X_{:,i}, i = 1$ to p **do**
- 3 **if** $\tilde{w}_i > w^*$ **then**
- 4 Add the feature $X_{:,i}$ to the set of high-weighting stressors I_{hw}
- 5 **end**
- 6 **end**
- 7 **Step 2:** Apply weights to the X matrix and compute weight-driven PCA
- 8 Transform the X matrix: $X_{weighted} = X \cdot \text{diag}(\tilde{w})$ and compute PCA on the weighted data matrix: $X_{weighted} = U\Sigma V^T$.
- 9 Keep the PCs that explain at least 5% variance, denote their loadings matrix $Q_{\tilde{w}} = V$ and the PC scores matrix $Z_{\tilde{w}} = U\Sigma$
- 10 **Step 3:** Remove the PCs that have top loadings on the minor-toxic stressors(earth elements)
- 11 **for** each loading vector Q_j **do**
- 12 Count the number $n_{le}^{PC_j}$ of earth elements in the top 9 loadings(1.5 times of the total number of earth element stressors)
- 13 **if** $n_{le}^{PC_j} \geq 4.5$ **then**
- 14 remove PC_j from the candidate PC set I_{pc} ; same for its loading vector Q_j
- 15 **end**
- 16 **end**
- 17 **Step 4:** Set PC-weights based on loadings on high-weighting stressors
- 18 For the left candidate PCs, set their PC-weights w_{pc_i} according to their **loadings** on the **high-weighting stressors**
- 19 **for** each loading vector $Q_j, j = 1, 2, \dots, \text{len}(I_{pc})$ **do**
- 20 Count the number $n_{hw}^{PC_j}$ of high-weighting stressors in the top $\text{len}(I_{hw})$ loadings
- 21 Assign the corresponding PC_j the weight as $w_{pc_j} = e^{1 + \frac{n_{hw}^{PC_j}}{\text{len}(I_{hw})}}$
- 22 **end**
- 23 **Step 5:** Compute pollution scores from the candidate PCs with PC-weights
Compute the pollution scores (SumReal) as the weighted sum of the qualified PC scores: $S_{\tilde{w}} = \sum_{i \in I_{pc}} w_{pc_i} \cdot Z_{\tilde{w}}[:,i]$
- 24 **return** pollution scores - $S_{\tilde{w}}(m, 1)$ and loadings of selected PCs - $Q_{sub|\tilde{w}}$

1.2 Sample Results Tables from the algorithms

Algorithm 2: Metrics of the completeness of the weight-driven PC loadings

Data: Standardized Chemical data matrix $X \in \mathbb{R}^{n \times p}$, weights $\tilde{w} \in \mathbb{R}^p$,
high-weighting threshold w^* ; fitted pollution scores: $S_{\tilde{w}}$, loadings of
selected PCs: $Q_{sub|\tilde{w}}$;

Result: Metrics for evaluating the completeness of the weight-driven PC loadings

- 1 **Step 1:** Prepare a baseline PCA results without stressor weights
 - 2 Apply PCA on the standardized data matrix X .
 - 3 Keep the PCs that explain at least 5% variance and do **not** remove the PCs with high loadings on earth elements
 - 4 Store the filtered baseline PCs matrix as Z_{base} and its loadings matrix as Q_{base}
 - 5 **Step 2:** Compute the similarity of each selected weight-driven loading vector to the corresponding baseline loading vector
 - 6 Select the equally important loadings in Q_{base} based on their rankings of eigenvalues
 - 7 **for** each loading vector Q_j in $Q_{sub|\tilde{w}}$ **do**
 - 8 **for** each loading vector Q_k in Q_{base} **do**
 - 9 **if** Q_k is equally important to the Q_j (same ranking number) **then**
 - 10 Compute the pearson correlation between Q_j and Q_k as $\rho_{j,k}$, add $\rho_{j,k}$ to the similarity set $\{\rho_{j,k}\}$
 - 11 Save Q_k as a column vector to produce $Q_{sub|base}$
 - 12 **end**
 - 13 **end**
 - 14 **end**
 - 15 **Step 3:** Compute the representativeness of the overall selected loadings in $Q_{sub|\tilde{w}}$
 - 16 **For** the two loading sets: $Q_{sub|\tilde{w}}$ and $Q_{sub|base}$, take the rotated data matrix XQ_{sub} and fit linear regression of X with the rotated matrix
 - 17 **for** each loading set $Q_{set} \in \{Q_{sub|\tilde{w}}, Q_{sub|base}\}$ **do**
 - 18 Fit linear regression models as $X = \mu + (XQ_{set})\beta + \epsilon$ and compute the R_{set}^2 using canonical correlations by: $R_{canonical}^2 = 1 - \frac{\det(E)}{\det(T)}$
 - 19 **end**
 - 20 Compute the ratio $\frac{R_{Q_{sub|\tilde{w}}}^2}{R_{Q_{sub|base}}^2} (\in [0, 1])$, it quantifies the representativeness of the weight-driven loadings data matrix X .
 - 21 **return** Similarity set $\{\rho_{j,k}\}$ and representativeness metric $\frac{R_{Q_{sub|\tilde{w}}}^2}{R_{Q_{sub|base}}^2}$
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Algorithm 3: Metrics of the discrimination ability of the weight-driven pollution scores for the high-weighting stressors

Data: The raw stressor data matrix $X_{raw} \in \mathbb{R}^{n \times p}$, the weight-driven pollution scores $S_{\tilde{w}}$, the baseline pollution scores S_{base} ;

Result: A metric of the discrimination ability on high-weighting stressors of the weight-driven pollution scores

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1 Step 1: Filter out the reference sites under two pollution scores
2 for each pollution scores  $S_{set} \in \{S_{\tilde{w}}, S_{base}\}$  do
3   Identify the reference sites as those with pollution scores lower than the 20-th
   percentile of  $S_{set}$ 
4   Store the filtered data matrix as  $X_{ref|set}$ 
5 end
6 Step 2: Make hypotheses tests for  $p$  times on the differences of stressor means
   between the two reference site groups  $S_{\tilde{w}}$  and  $S_{base}$ 
7 for each stressor feature  $X_i, i = 1, 2, \dots, p$  from the reference site groups do
8   if  $X_i$  is a high-weighting stressor then
9     Take a conservative estimate of the difference of means as
        $\hat{D}_{mean,i} = (\bar{X}_{upper95\%,i|\tilde{w}} - \bar{X}_{lower5\%,i|base})$ 
10    Set the null hypothesis  $H_0 : D_{mean,i} \geq 0$ , compute the t-statistic, and
       p-value;
11    Add the p-value to the p-value set  $\{p\text{-value}_i\}_{hw}$ , set the critical level as 0.01
       for the p-value to reject  $H_0$ 
12  end
13  if  $X_i$  is not a high-weighting stressor then
14    Take a conservative estimate of the difference of means as
        $\hat{D}_{mean,i} = (\bar{X}_{upper95\%,i|\tilde{w}} - \bar{X}_{lower5\%,i|base})$ 
15    Set the null hypothesis  $H_0 : D_{mean,i} = 0$ , compute the t-statistic and
       p-value; Add the p-value to the p-value set  $\{p\text{-value}_i\}_{lw}$ , set the critical
       level as 0.05 for the p-value to reject  $H_0$ 
16  end
17 end
18 Step 3: Compute the group mean differences within weight-driven reference sites
19 Partition the sites into three groups:  $X_{ref|\tilde{w}}$ ,  $X_{med|\tilde{w}}$  and  $X_{degr|\tilde{w}}$  based on the
   20-th, 80-th percentiles of the weight-driven pollution scores  $S_{\tilde{w}}$ 
20 Do PERMANOVA test to assess whether the raw stressor means differ among the
   three groups
21 Set the null hypothesis  $H_0 : \mu_{ref} \geq \mu_{med} \geq \mu_{degr}$ 
22 Compute Pseudo-F statistic:  $Pseudo - F = \frac{SS_{between}/(k-1)}{SS_{within}/(N-k)}$ ,  $p\text{-value} = \frac{\#(F_{perm} \geq F)}{N_{perm}}$ 
23 return The t-test p-value sets  $\{p\text{-value}_i\}_{hw}$ ,  $\{p\text{-value}_i\}_{lw}$  and the PERMANOVA
   test results - Pseudo-F statistic and p-value

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As	Weight	Similarity set	representativeness ratio
	3	(0.71, -0.47)	0.88
	10	0.15	0.61
	1	(1, 1, 1, 1)	1

Table 1: Sample table with 3 rows and 4 columns

Variable	Comparison	D _{mean}	t	p	Significance
As	HW (3)	-0.1232	-2.7709	0.0042	**0.01**
As	HW (5)	-0.2184	-3.1730	0.0014	**0.01**
As	HW (10)	-0.3878	-4.0259	0.0001	**0.01**
%OC	LW (3)	1.0705	2.4612	0.0183	*0.05*
1234-TCB	LW (3)	1.1662	1.2441	0.2208	ns
1245-TCB	LW (3)	0.4792	0.5332	0.5969	ns
Al	LW (3)	0.5110	2.4239	0.0197	*0.05*
%OC	LW (5)	0.9888	2.2855	0.0277	*0.05*
1234-TCB	LW (5)	1.1877	1.3084	0.1981	ns
1245-TCB	LW (5)	0.4862	0.5790	0.5580	ns
Al	LW (5)	0.5101	2.4172	0.0203	*0.05*
%OC	LW (10)	1.0098	2.3822	0.0221	*0.05*
1234-TCB	LW (10)	1.2679	1.6239	0.1123	ns
1245-TCB	LW (10)	0.4428	1.5214	0.1360	ns
Al	LW (10)	0.5121	2.7039	0.0100	*0.05*